

TMUA Mock Paper

ZVCH

75 minutes

Question 1

Let a_n be the area bounded by the curves $x = y$, $x^2 + y^2 = n$ and the positive x-axis. What is $\sum_{n=1}^{100} a_n$?

A: $\frac{250\pi}{2}$

B: $\frac{50\pi}{4}$

C: $\frac{5050\pi}{8}$

D: $\frac{3805\pi}{8}$

E: $150\sqrt{2}\pi$

Question 2

The symbol $\{x\}$ denotes the fractional part of x . For example $\{10.4\} = 0.4$, $\{2\} = 0$. Given that $f(x) = 5x^3 + 2x^2 + 8x + 1$, find

$$\int_0^n f(\{x\} + 1) dx$$

where n is a positive integer.

A: $\frac{425n}{13}$

B: $\frac{1}{2}n(n+1)$

C: $\frac{83n}{12}$

D: $\frac{437n}{12}$

E: $15n^2 + 4n + 8$

Question 3

Let x and y be real numbers such that $x^2 + y^2 = 49$. What is the greatest value of $12x - 5y$?

- A: 91
- B: 84
- C: 104
- D: 89
- E: 87

Question 4

For what values of x is

$$x + 3 = 1 + 1 + x + 1 + 2x + x^2 + 1 + 3x + 3x^2 + x^3 + \dots$$

A: -1

B: 0

C: $\frac{-3-\sqrt{5}}{2}$

D: $\frac{-3+\sqrt{5}}{2}$

E: $\frac{-3-\sqrt{5}}{2}$ and $\frac{-3+\sqrt{5}}{2}$

Question 5

How many four-digit square numbers are multiples of 3, 4 and 5?

A: 0

B: 1

C: 2

D: 3

E: 4

F: 5

G: 6

Question 6

Two identical ice sculptures are left to melt outside. Sculpture A melts in 6 hours. Sculpture B melts in 12 hours. Assuming each sculpture melts uniformly, after how long will Sculpture B have twice as much ice remaining as Sculpture A?

- A: 3 hours
- B: 3.5 hours
- C: 4 hours
- D: 4.5 hours
- E: 5 hours

Question 7

Alvin and Simon can create a song in 5 days. Alvin and Theodore can create a song in 8 days. Simon and Theodore can create a song in 10 days. Approximately how many days does it take Alvin, Simon and Theodore to make a song together?

A: 3

B: 3.2

C: 3.5

D: 3.7

E: 4.7

Question 8

Find x such that

$$\ln(x \ln(x \ln(x \ln(x \ln(x \cdots = e$$

A: e

B: e^3

C: e^{e+1}

D: e^e

E: e^{e-1}

Question 9

Find the coefficient of x^{15} in the expansion of $(2x^3 + 3x + 4x^5)^5$.

A: 11280

B: 9920

C: 8462

D: 12482

E: 10592

Question 10

Find the units digit of $34657^{267} + 81$.

A: 0

B: 1

C: 2

D: 3

E: 4

F: 5

G: 6

H: 7

I: 8

J: 9

Question 11

There are 129 coins on a table. 128 are fair, and one has heads on both sides. A coin is tossed 8 times and lands on heads every time. What is the probability it lands on heads if thrown a 9th time?

A: $\frac{1}{2}$

B: $\frac{9}{10}$

C: $\frac{7}{8}$

D: $\frac{5}{6}$

E: $\frac{2}{3}$

Question 12

Find the sum of the solutions to

$$16^{\log_2 x} - 10 \times 27^{\log_3 x} + 35 \times 64^{\log_8 x} - 50x + 24 = 0$$

- A: 0
- B: 8
- C: 10
- D: 24
- E: 43

Question 13

How many solutions does

$$\sin x + \sin^2 x + \sin^3 x + \cdots = \tan x$$

have for $0 \leq x \leq 2\pi$?

A: 1

B: 2

C: 3

D: 4

E: 5

F: 6

Question 14

How many sets of positive integers (a, b, c) with $a \leq b \leq c$ satisfy

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1?$$

- A: 1
- B: 2
- C: 3
- D: 4
- E: 0
- F: ∞

Question 15

Given

$$f(x) = \frac{x\sqrt{x} - 2\sqrt{x}}{\sqrt{2} - \sqrt{x}},$$

find $f'(2)$.

A: 0

B: $\frac{1}{\sqrt{2}}$

C: $\frac{1}{2}$

D: $-\frac{3}{2}$

E: $\frac{3}{2}$

Question 16

A sheep called Shaun lives in a circular pen of radius 8. His position inside the pen is random. What is the probability that Shaun is within 1m of a side of the pen?

A: $\frac{1}{2}$

B: $\frac{49}{64}$

C: $\frac{7}{8}$

D: $\frac{15}{64}$

E: $\frac{1}{8}$

Question 17

Solve:

$$\log_{x-2} 9 = \log_x 81$$

A: $x = 1$

B: $x = 4$

C: $x = 1$ and $x = 4$

D: No real solutions

E: All of the above

Question 18

For which function is

$$f(f(f(f(f(x)))))$$

never equal to 0 for any real x ?

A: $f(x) = \sin(x)$

B: $f(x) = x^7$

C: $f(x) = e^x - 1$

D: $f(x) = \cos^2(x)$

Question 19

A proper divisor of a number is a positive factor that is smaller than the number.

E.g. The proper divisors of 6 are 1,2,3

How many proper divisors does 30^4 have?

A: 63

B: 125

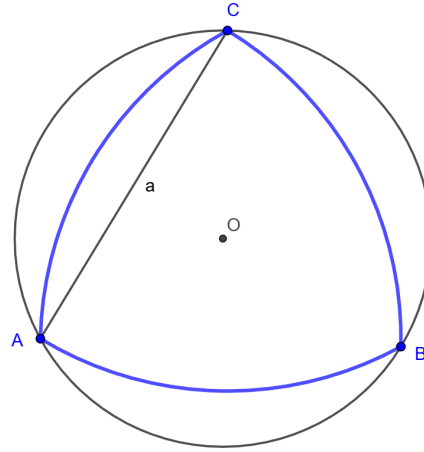
C: 124

D: 64

E: None of the above

Question 20

Given that CB is an arc formed from the circle of radius a centred at A, and AB and AC are defined similarly, what is the area of the shape ABC in terms of a ?



A: $\frac{a^2}{2} \left(\pi - 3 \tan \frac{\pi}{6} \right)$

B: $\frac{a^2}{2} \left(\pi - 3 \tan \frac{\pi}{3} \right)$

C: $\frac{a^2}{2} \left(\pi - 3 \sin \frac{\pi}{6} \right)$

D: $\frac{a^2}{2} \left(\pi - 3 \sin \frac{\pi}{3} \right)$

E: $3a^2 \pi \tan \left(\frac{\pi}{6} \right)$

Answer Key

1. C
2. D
3. A
4. D
5. C
6. C
7. E
8. E
9. E
10. E
11. D
12. C
13. C
14. C
15. D
16. D
17. B
18. D
19. C
20. A